Pupil Labs Neon Eye-tracking Manual: Boston College Human Neuroscience Lab

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Table of contents

Code of conduct 2
- Where can it be used? 2
- **System/Hardware Overview** 2
  - **Components of the Neon System** 2
- Who can use it? 3
  - Roles 4
- Support/Training 4
- Booking 4
- Costs 5
- Data privacy 5
- File Storage 5
- Cleaning 6
- DOs and DON'Ts 6
  - Dos 6
  - Don'ts 6

User Guide 6
- Before the Participant Arrives 6
- Upon Participant Arrival 7
  - Eyeglasses 7
  - Contact lenses 7
  - Makeup 7
  - Gaze validation 8
  - Offset correction 8
- Live Streaming 8
- Recording 9
- After the Participant Leaves 10
- Data Export 10
- Troubleshooting 11
- Session Checklists 12
  - Before the participant arrives 12
  - With the participant 12
  - After the participant leaves 12
- Sources 12

Code of conduct

The Human Neuroscience Laboratory houses six Pupil Labs Neon wearable eye-trackers.

This manual is the standard way to set up, fit, record, and offload data with the Neon. Follow it for every session so data are comparable across studies and RAs.

- **Covers**: equipment care, pre-session checks, participant fitting, recording, upload, and data handling.
- **Does not cover**: study-specific task scripts, stimulus presentation, or analysis pipelines. Those live in each study's protocol.
- Each study's IRB-approved protocol overrides this manual where they conflict. For screen-based tracking, see the Tobii Pro Spectrum manual.

Where can it be used?

The Neon is portable and can be used in the lab or in the field. Devices are checked out through iLab (see Booking) and must be returned to the HNL after each reservation.

System/Hardware Overview

- Neon is a calibration-free wearable eye-tracker: a sensor module in a glasses frame, tethered by USB-C to an Android Companion phone that runs gaze estimation and stores recordings.
- The module contains binocular infrared eye cameras (850 nm), a front-facing scene camera, a microphone, a 9-DoF IMU, and indicator LEDs.
- Gaze is computed in real time on the Companion phone. Recordings can be processed in Pupil Cloud or offline in Neon Player.

> ! Important
>
> Nothing is recorded unless the phone is connected and the Neon Companion app is running.

Components of the Neon System

Component	What it does	Lab notes
Neon module	Binocular IR eye cameras, scene camera, microphone, IMU, indicator LEDs	Never touch camera lenses with fingers. Serial number is on the back (Data Matrix).
Frame (nest)	Holds the module; attached with two screws	Record which frame was used in the run log. Swap frames only per the swapping guide.
USB-C cable	Powers the module and carries data to the phone	Main failure point. Avoid sharp bends; route behind the wearer.

Component	What it does	Lab notes
Companion phone	Runs the Neon Companion app; computes gaze in real time; stores recordings	Lab phone model: Samsung Galaxy S25.
Neon Companion app	Wearer profiles, recording, events, offset correction, upload	Update from the Play Store only between studies, never mid-study.
Prescription lens frame	Frame with swappable corrective lenses for participants who wear glasses	The lab has one, with lenses from −3 to +3 diopters. Only one is available, so check it is free before scheduling participants who need it.
Neon Monitor (browser)	Live view of scene + gaze from a laptop on the same network	Used by the experimenter to check data quality during sessions.
Pupil Cloud	Cloud storage, gaze/fixation processing, enrichments, export	Only if approved by the IRB for the study (see Data privacy).
Neon Player	Offline desktop app for viewing and exporting recordings	Use when data cannot go to the cloud.

> ! Important
>
> Do not install Android system updates on the Companion phone unless the version is on Pupil Labs' supported list. Unsupported versions can break USB access to the module.

Additional Notes:

- **Data Management**: Please transfer recordings off the phone as soon as possible (see After the Participant Leaves).
- **Contact**: Contact Jason Geller when something is missing, broken, or when supplies are running low.

Who can use it?

- Contact the director for the rules, regulations, and access to the booking calendar.
- The Neon trackers are available to BC faculty, students, and external users.
- Usage is not free [Costs].
- To use the system, you must complete training with the director.
- Only trained, IRB-listed personnel may run Neon sessions.

Roles

Role	Responsibilities
Lab director	Approves this manual, Pupil Cloud use, and workspace membership
Lab manager / equipment lead	Owns the hardware, phone OS and app versions, workspace admin, and this manual's revisions
Study lead (grad student / postdoc)	Confirms study-specific settings, wearer naming, templates, and export plan
RA / experimenter	Runs sessions per this manual, keeps the run log, reports problems the same day

Support/Training

- **Training before running participants alone**:
  1. Read this manual and the study protocol.
  2. Complete current CITI human subjects training and be listed on the IRB protocol.
  3. Shadow at least two sessions with a trained experimenter.
  4. Run two pilot sessions under supervision (one on a labmate), including upload and log entry.
  5. Sign off with the lab manager.
- **Note**: Support generally does not include programming your experiment for you.
  - A PsychoPy template is available to help you program your experiment: https://github.com/jgeller112/PL_Demo
  - Troubleshooting assistance is provided if the experiment is not working properly.
  - **Preprocessing/Analyzing Data**:
    - Tutorials are available to help with data preprocessing and analysis.
    - Consulting is an additional cost.
    - Helping you design your study, analyze data, or performing analyses/write-up results contributes to your project under the credit taxonomy https://credit.niso.org/ and warrants authorship (regardless of payment).

Booking

- Book the Neon via iLab (email John Maloney [malonegl@bc.edu] for access). Ensure that the booking is under the appropriate schedule.
  - Pupil Labs Neon eye-trackers
- Before starting a new study, please fill out this Qualtrics form: https://docs.google.com/forms/d/e/1FAIpQLScizVkYkFd064MXpGUShDfyR9XHXktQKEyPyIgszIwggkoAbw/viewform
- **Every new lab and every new study must set up its own Pupil Cloud workspace.** Do not record into another lab's or study's workspace. Contact the lab manager to have the workspace added to the Companion app.

> ! Important
>
> Recordings are stored in whichever workspace is selected in the Companion app when you record. Always select your study's workspace in the app before recording.

Costs

- You pay for each hour of use:
  - **Self Use** Mon-Fri 12:00 AM - 05:00 PM (Internal: $6.00/hr, External Non-Academic: $30.00/hr, External Academic: $30.00/hr)
  - **Assisted Use** Mon-Fri 09:00 AM - 05:00 PM (Internal: $102.92/hr, External Non-Academic: $199.41/hr, External Academic: $199.41/hr) (needs approval)
  - **Training Session** Mon-Fri 09:00 AM - 05:00 PM (Internal: $102.92/hr, External Non-Academic: $199.41/hr, External Academic: $199.41/hr) (needs approval)

Data privacy

Scene video captures faces, screens, and audio of anyone nearby, so it is identifiable data from the moment recording starts.

- Confirm the consent form covers video, audio, and (if used) cloud storage on third-party servers before any recording.
- Turn audio recording off in the Companion app unless the protocol requires it.
- Never record bystanders who have not consented. In shared spaces, point out recording to others or pause.
- Use participant IDs only. Never put names, initials, or birthdates in wearer profiles, recording names, or events.
- Do not copy recordings to personal devices or personal cloud accounts.

File Storage

- **Naming**: `STUDYCODE-PID_S#` for recordings (e.g., `MWC-P012_S1`), matching the wearer name. One Pupil Cloud project per study.
- Raw recordings are kept unchanged as the source of truth; every analysis starts from an export, never from edited raw files.
- **What to keep**:
  1. The native recording folder (never modified), on secure lab storage with a backup.
  2. The CSV export (gaze, fixations, saccades, blinks, events, IMU, and eye states where available) in the study's `data/raw/` folder.
  3. The run log, saved alongside the data.
  4. A note of the Companion app version, Pupil Cloud or Neon Player version, and any enrichment settings.
- **Sharing**: Scene videos are identifiable and are not posted publicly. Share de-identified CSVs and code per the study's data-sharing plan.

> ! Important
>
> Phone storage is limited. Delete recordings from the phone only after the cloud or server copy is verified.

Cleaning

- Wipe scene and eye camera lenses gently with the supplied microfiber cloth only, and only if dusty. Never use liquid on the module.
- Disinfect the frame (nose pads, temples) with an alcohol disinfectant wipe before and after every participant, and let it dry fully.
- Do not let wipes touch the cameras or USB connectors.

DOs and DON'Ts

Dos

- **Charge the Companion phone to 100% before every session** (overnight with the supplied fast charger).
- Log every recording in the run log, including which frame and module were used.
- Report hardware damage, failed recordings, or app errors to the lab manager the same day.
- Always book all sessions via iLab. This includes booking time to pilot your experiment.
- **Always start your session in the iLab kiosk before beginning, and end it in the kiosk when you are done.**

Don'ts

- Don't install Android system updates or update the Neon Companion app mid-study.
- Don't touch camera lenses with your fingers or use liquid on the module.
- Don't sharply bend or tug the USB-C cable.
- Don't leave your data on the phone. Recordings must exist in at least two places before anything is deleted.

User Guide

Before the Participant Arrives

Start about 30 minutes before the participant arrives. The goal is a charged, connected, clean system with enough storage and the right study settings.

1. **Start your session in iLab.** Always start your session in the iLab kiosk before beginning. Navigate to `Go to Kiosk` under scheduling, find your reservation, and start your session.
2. **Charge.** Companion phone at 100%. For sessions over about 3 hours, use a powered USB-C hub to charge during recording.
3. **Inspect hardware.** Check the module screws are tight, the cable has no kinks or frayed ends, and connectors are clean and dry.

4. **Clean and disinfect** (see Cleaning).
5. **Check the phone.**
   - Airplane mode or Do Not Disturb on; Wi-Fi on only if using Neon Monitor or uploading.
   - Free storage at least [fill in, e.g., 20 GB].
   - Neon Companion app version matches the version logged for this study.
   - Android version is on the supported list. Decline any system update prompts.

   > **i** Note
   >
   > Roughly 1 GB per 10 minutes of recording is a safe planning figure; confirm with a test recording.

6. **Check workspace and settings.** In the app, select your study's workspace (not another lab's or study's) and confirm it is active, audio is set per protocol, and cloud upload is on or off per protocol.
7. **Connect and test.** Plug the cable into the phone and wait for the app to recognize Neon. Put the glasses on yourself, start a 30-second test recording, and confirm the gaze circle tracks your fixations. Delete the test recording.
8. **Set up the space.** Remove or cover identifying material from the scene camera's view (posters with names, open screens). Prepare the run log (paper or [fill in: shared sheet location]).
9. **If streaming:** set up the live stream (see Live Streaming) and confirm the live scene and gaze appear.

Upon Participant Arrival

> **i** Difficulties with eyeglasses, makeup, etc.
>
> **Eyeglasses**
> Regular glasses usually cannot be worn under Neon. The lab has **one prescription lens frame** with lenses from **-3 to +3 diopters**. Participants whose prescription is outside this range should wear contacts or be handled per the study's exclusion rules.
>
> **Contact lenses**
> Contacts are fine.
>
> **Makeup**
> Heavy mascara or eyeliner can reduce eye-image quality. Note it in the run log.

Neon needs no calibration, but every participant gets a fit check and a gaze validation before the task. This takes about 5 minutes.

1. **Obtain consent** per the IRB protocol, including video, audio (if on), and cloud storage (if used). Do not put glasses on before consent.
2. **Screen for fit issues.** Ask about prescription glasses, contact lenses, and heavy eye makeup (see above). If the participant needs the prescription frame, ask for their prescription and fit the matching lenses (-3 to +3 only).
3. **Create the wearer profile.** In the Companion app, add a new wearer named with the participant ID only (e.g., `STUDY-P012`). Use one profile per participant so offset corrections never carry over to the wrong person.
4. **Enter IED if the study uses pupil size.** Measure inter-eye distance and enter it in the wearer profile.
5. **Fit the glasses.** Seat them like normal glasses: centered, resting on the nose and ears, not tilted.
   - Attach the head strap for any task with head movement.
   - Route the cable behind the head and clip the phone to the participant's waistband or place it on the table.
6. **Check eye images.** In the app, both eyes should be fully visible in the eye video. If not, reseat the glasses or try a different nose pad.

Gaze validation

Ask the participant to look at 3 to 5 known targets (e.g., the corners and center of a target card at the task distance). The gaze circle should land on each target.

Offset correction

Apply offset correction only if gaze is consistently shifted in one direction. Do it in the wearer profile while the participant fixates a target at the task's viewing distance.

> **i** Note
>
> Offset is distance-specific, so close-range tasks (under about 1 m) are most affected.

Re-validate after any correction and log the result (**pass** / **pass after correction** / **fail**).

Live Streaming

You can watch the scene video and gaze live on a computer with Neon Monitor.

1. **Connect the phone and the computer to the same Wi-Fi network.** The live stream only works when both devices are on the same network.
2. Open the Stream screen in the Companion app to see the phone's address.
3. Enter that address in a web browser on the computer. The live scene and gaze should appear.

> **Note**
>
> If nothing appears, campus Wi-Fi may be blocking the connection between devices. Use a dedicated lab router or hotspot instead (see Troubleshooting).

Recording

Start the recording before the task begins and stop it only after the task ends, so no trial is lost at the edges. Mark every task boundary with an event.

1. **Start recording.** Press the record button in the Companion app (or start it from the task script via the Real-Time API). Confirm the timer is running.
2. **Record a validation segment.** Have the participant fixate the target card points for about 5 seconds each. This lets analysts check accuracy post hoc.
3. **Send events.** Mark task start, block boundaries, and task end. Use the study's event names exactly (e.g., `task_start`, `block1_start`).

   > **Note**
   >
   > Events can be added from the app, Neon Monitor, or the experiment script; scripted events are preferred for timing precision.

4. **Monitor during the session.** Watch Neon Monitor or glance at the phone between blocks. Stop and fix if:
   - the phone vibrates or the module's red LED blinks (low battery, low storage, or connection loss; the app shows the reason);
   - the gaze circle disappears or is clearly off target;
   - the glasses have shifted noticeably on the face.
5. **After a refit mid-session**, re-run the validation segment and add an event (`refit`) so analysts can separate the data.
6. **End the task.** Send the `task_end` event, re-run the validation segment, then stop the recording.
7. Recording from Multiple Glasses

The Companion app controls one pair of glasses at a time. To record from more than one pair at once (e.g., several participants together), use one of these tools instead of starting each phone by hand:

- **SocialEyes**: streams, records, and synchronizes data from multiple Neon devices. https://github.com/beatlab-mcmaster/socialeyes
- **Pupil Labs TUI**: a terminal interface for controlling several Neon devices at once.

> **Note**
>
> Each pair of glasses still needs its own Companion phone, wearer profile, and run log entry. All phones must be on the same network as the computer running the tool.

After the Participant Leaves

The recording is not safe until it exists in at least two places. Finish these steps before the next participant.

1. **Confirm the recording saved.** Open it in the app's recordings list, play a few seconds, and check that gaze is overlaid.
2. **Remove the glasses.** Unplug the cable from the phone first, then remove the glasses from the participant.
3. **Upload or offload.**
   - **Cloud studies**: connect to Wi-Fi and confirm the upload completes (the cloud icon shows done) before deleting anything from the phone.
   - **Offline studies**: connect the phone to the lab computer by USB, copy the recording folder to the study's secure storage location, and verify the copy opens in Neon Player.
4. **Clean** (see Cleaning).
5. **Store.** Put the glasses in their case, coil the cable loosely, and put the phone on the charger.
6. **Delete from the phone** only after the cloud or server copy is verified, per the study's data plan.
7. **Report problems** (hardware damage, failed recordings, app errors) to the lab manager the same day.
8. **End your session in iLab.** Return to the iLab kiosk and end your session.

Data Export

Route	When to use	Steps
Pupil Cloud	IRB permits third-party cloud storage	Add recordings to the study project, run enrichments (e.g., Marker Mapper for screens, Reference Image Mapper for scenes) as specified by the study lead, then download the Timeseries Data + Scene Video export.
Neon Player	Data must stay on lab servers	Open the native recording folder copied from the phone,

Route	When to use	Steps
		check gaze and events, then export CSVs and video.

> ! Important
>
> Post-hoc offset correction is possible in Pupil Cloud, but it changes all downstream data (fixations, mapped gaze). Only the study lead applies it, and the change is logged.

Troubleshooting

Most problems are the cable, the phone's battery or storage, or an Android update. Check those first; full guidance is in Pupil Labs' troubleshooting documentation.

Symptom	Likely cause	Fix
Phone vibrates, red LED blinks on module	Low battery, low storage, or unstable connection	Open the app to read the error. Charge, free space, or reseat the cable.
App does not recognize Neon / cameras not shown	Loose cable, wrong cable, or unsupported Android version	Reseat both cable ends; restart the app; check Android version against the supported list.
Android updated by accident	Unsupported OS breaks USB access	Stop using the phone for data. Tell the lab manager; a rollback erases the phone, so back up first.
Gaze consistently shifted	Parallax at close range or wearer physiology	Reseat glasses, re-validate, then apply offset correction at task distance.
Gaze jumps or drops out	Eye not fully in view, glasses slipping, strong sunlight	Reseat glasses, add head strap, reduce direct sunlight on the eyes.
Neon Monitor shows nothing	Laptop and phone on different networks, or campus Wi-Fi blocks device discovery	Use a dedicated lab router or hotspot; enter the phone's IP from the app's Stream screen.
Upload stuck	No Wi-Fi or large file	Keep the phone on Wi-Fi and charging; do not delete the local copy.

If none of these work, stop the session, log the issue, and contact the lab manager. Pupil Labs support: Discord or info@pupil-labs.com (have the module serial number ready).

Session Checklists

Print and keep with the device.

Before the participant arrives

- Session started in iLab kiosk
- Phone at 100%, storage above minimum
- Cable, screws, connectors inspected
- Frame disinfected and dry; lenses clean
- Study's own workspace selected in the app; audio and upload settings correct
- Self-test recording done and deleted
- Room cleared of identifying material; run log ready

With the participant

- Consent signed (video, audio, cloud as applicable)
- Wearer profile created with participant ID only
- Glasses fitted; both eyes visible
- Gaze validated; offset applied only if needed
- Recording started; validation segment recorded
- Events sent at task start, blocks, and end
- Closing validation segment; recording stopped

After the participant leaves

- Recording plays back with gaze
- Uploaded or copied to secure storage and verified
- Run log completed
- Frame disinfected; device cased; phone charging
- Problems reported to lab manager
- Session ended in iLab kiosk

Sources

Pupil Labs Neon documentation (https://docs.pupil-labs.com/neon/), accessed 2026-10-07:

- Neon module technical overview
- Compatible Companion devices and Android versions
- Cleaning and disinfecting
- Wearers and IED
- Offset correction in the Companion app
- Offset correction in Pupil Cloud
- Troubleshooting